1.Aim of the practical: Implementation of Python Basic Libraries such as
Math
NumPy
Matplotlib
Seaborn
SciPy.
2. Creation and loading different Datasets in Python.
3. Write a program to implement Linear regression.
Download different datasets from Kaggle or UCI ML Repository.
Import all the necessary modules for the datasets.
Find out the best fit line along with the MSE, RMSE.

2.Objective: To develop the ability to create, load, and manipulate various datasets in Python using Jupyter Notebookwith essential Python libraries — Math, NumPy, Matplotlib, Seaborn, and SciPy — which are foundational tools for data analysis, scientific computing, and data visualization.

3.Algorithm/Flowchart:

```
import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error

data = pd.read_csv('insurance.csv')

print("Mean of age:", data['age'].mean())
print("Median of age:", data['age'].median())
print("Mode of age:", data['age'].mode())

plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
data['age'].hist(color='skyblue', edgecolor='black')
plt.title('Histogram of Age')

plt.subplot(1, 2, 2)
data['bmi'].hist(color='salmon', edgecolor='black')
plt.title('Histogram of BMI')
```

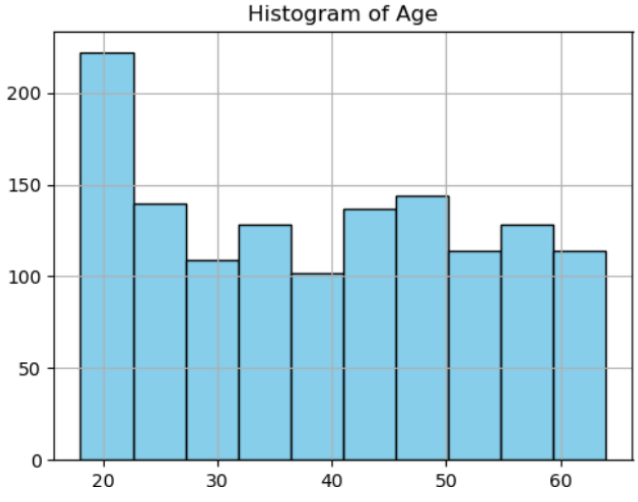
```
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
plt.scatter(data['age'], data['bmi'], color='green', edgecolors='black', marker='o')
plt.title("Scatter Plot of Age vs BMI")
plt.xlabel("Age")
plt.ylabel("BMI")
plt.grid(True)
plt.show()

X = data[['age', 'bmi']]
y = data['charges']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)

print("\nLinear Regression Results:")
print("MSE:", mse)
print("RMSE:", rmse)
print("Coefficients:", model.coef_)
print("Intercept:", model.intercept_)

from mpl_toolkits.mplot3d import Axes3D
fig = plt.figure(figsize=(10,7))
ax = fig.add_subplot(111, projection='3d')
ax.scatter(X_test['age'], X_test['bmi'], y_test, c='blue', label='Actual Charges')
ax.scatter(X_test['age'], X_test['bmi'], y_pred, c='red', label='Predicted Charges', alpha=0.5)
ax.set_xlabel('Age')
ax.set_ylabel('BMI')
ax.set_zlabel('Charges')
plt.title("3D View: Age, BMI vs Charges")
ax.legend()
plt.show()
```

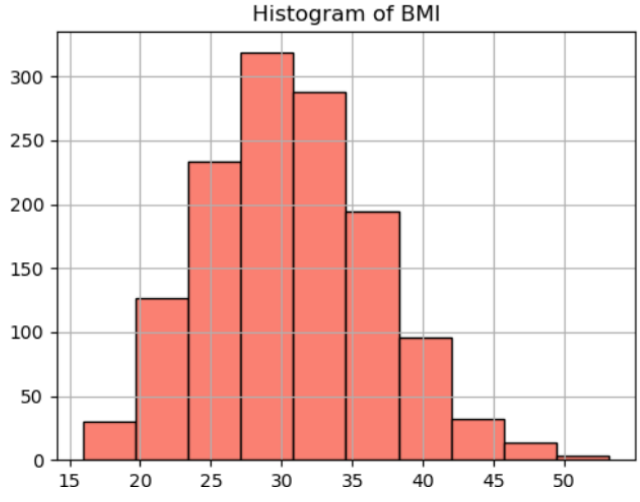
4.Result/Output:

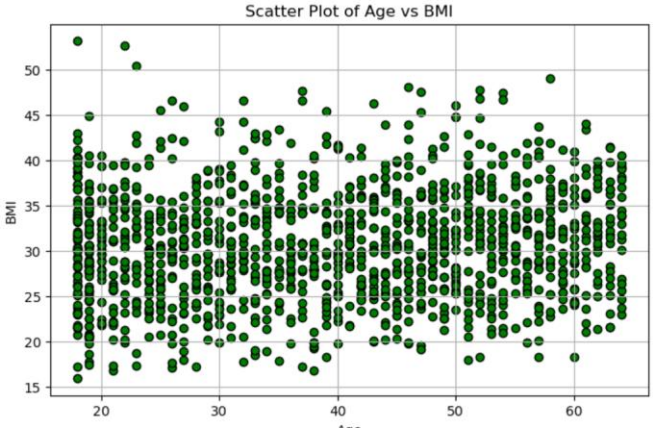
```
Mean of age: 39.20702541106129
Median of age: 39.0
Mode of age: 0    18
Name: age, dtype: int64
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520
...	...	...	...	...	...	...	...
1333	50	male	30.970	3	no	northwest	10600.54830
1334	18	female	31.920	0	no	northeast	2205.98080
1335	18	female	36.850	0	no	southeast	1629.83350
1336	21	female	25.800	0	no	southwest	2007.94500
1337	61	female	29.070	0	yes	northwest	29141.36030

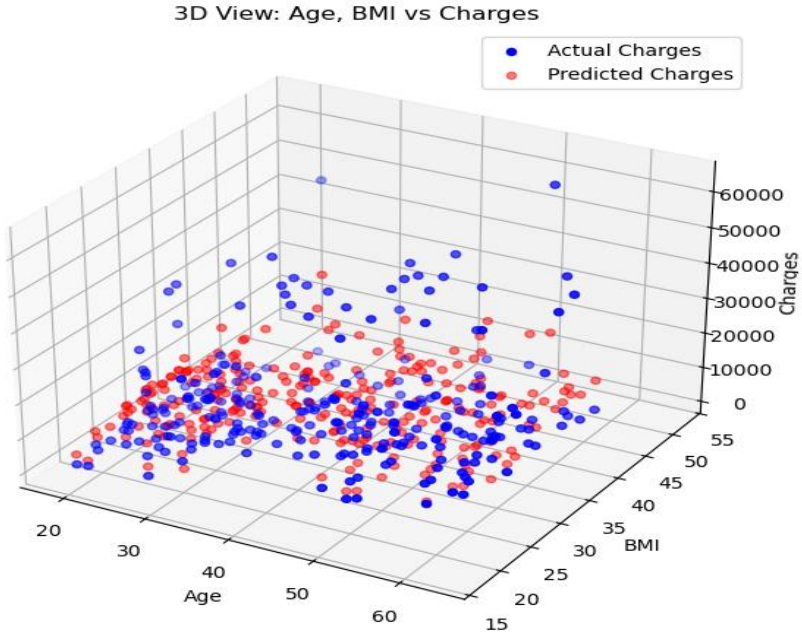
1338 rows × 7 columns







```
Linear Regression Results:
MSE: 131440262.76073726
RMSE: 11464.739977894713
Coefficients: [223.79653146 330.79448868]
Intercept: -5571.083408548651
```



Learning outcomes (What I have learnt):

- Load datasets from CSV, Excel, or built-in sources using Python tools like pandas, NumPy, or open() functions.
- Handle basic arithmetic and scientific calculations.
- Plot line charts, bar graphs, histograms, and scatter plots.